

Fares Mohammed Nabil Elsherif

Giza, Egypt | 01114101919 | s-fares.nabil@zewailcity.edu.eg | [Github](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Communications & Information Engineering student at Zewail City of Science and Technology, with proven hands-on experience designing, building, and debugging hardware/software integrated control systems. Demonstrated ability to engineer real-time closed-loop systems, industrial circuits, and sensor-actuator pipelines that deliver measurable results — from eliminating timing drift to cutting manual intervention to zero. Deeply motivated by industrial automation and mechatronics, actively seeking an internship to contribute immediately and develop professional-grade engineering skills.

TECHNICAL SKILLS

Languages: C++ (OOP, STL), Python (GUI, scripting)

Machine Control: Machine control software, closed-loop & PID control, real-time systems

Automation control: Sensor-actuator integration, automated control, process automation

Circuit Design: Industrial circuit design, NE555, CD4017, I2C, amplifiers, voltage regulation

Embedded Systems: Arduino, ESP32, ESP32-CAM, MPU6050 IMU, ESP-NOW, Blynk IoT dashboard, edge-based signal detection

Tools & Languages: Git, GitHub, signal processing, documentation | Arabic (native), English (professional)

PROJECTS

MAWZOON — Self-Balancing Robot | *ESP32 · MPU6050 · L298N · PID Control · C++*

- Designing a 2-wheeled self-balancing robot on ESP32, using a 6-axis MPU6050 IMU (I2C) for real-time tilt sensing that feeds a closed-loop PID balance controller.
- Engineered the power architecture — 2S (7.4V) 18650 Li-ion pack with a 470–1000μF decoupling capacitor on the 5V rail — paired with an L298N dual H-bridge driving 2 DC geared motors via independent PWM channels.

BEITI-TECH — AI Smart Home System | *ESP32 · ESP32-CAM · ESP-NOW · Blynk · C++*

- Built a dual-ESP32 system (main unit + ESP32-CAM) linked over ESP-NOW; implemented ESP-DL face recognition that auto-unlocks the door for known residents and sends Telegram alerts for unrecognized visitors.
- Integrated 6 sensor/actuator modules (DHT11, PIR, MQ-2 gas, flame, servo lock, buzzer) into a Blynk dashboard for automated lighting/fan control and gas/fire safety alerts; presented as Team 15 at Zewail City's 3rd Innovation Exhibition.

AQUASMART — IoT Smart Fish Farm Monitoring System | *ESP32 · Blynk · C++*

- Engineered a real-time water-quality system on ESP32 integrating 5 sensors (pH, $\pm 0.5^{\circ}\text{C}$ DS18B20 temperature, water level, YF-S201 flow, and a derived dissolved-oxygen reading) with SD-card CSV logging and a Blynk dashboard.
- Derived a novel temperature/flow-based dissolved-oxygen equation, cutting hardware cost to ~1,200 EGP and removing the need for a dedicated DO sensor; validated over a 7-day, 60L trial (pH 7.0–8.2, DO 4.8–6.2 mg/L) with automatic pump control.

Electronic Circuit Design Portfolio | *Industrial Circuit Design · Debug · Documentation*

- Designed, built, and debugged 5 functional circuits (DC power supply, traffic light controller, diode logic gates, amplifier); resolved 3+ design faults through fault isolation; produced a full annotated documentation portfolio.

EDUCATION

Zewail City of Science and Technology — B.Sc. Communications & Information Engineering

2025 – 2029

Coursework: Calculus, Physics, OOP, Electronic Circuits, Engineering Design

El-Sadat STEM High School — STEM Track | 89%

Graduated 2025

UNIVERSITY ACTIVITIES & CERTIFICATES

ZC Ambassador Team Member, Zewail City Career Fair 2026

- Guided over **160+ attendees** and managed on-site logistics across **30+ company booths** during the career fair event.

IEEE Student Branch

- Participated in **5+ embedded systems workshops** as an active student member, collaborating in small teams of **4-6 members** to prototype and troubleshoot hardware-software solutions.

Digital Egypt Cubs Initiative (DECI)

- Completed an intensive **90-hour** training track, earning a professional certification ,2024